Mark schemes

Q1.

(a) $\frac{dx}{dt} = 3t^2$ $\frac{dy}{dt} = 12t$

both correct

B1

$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 9t^4 + 144t^2$

'their' $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2$

M1

$s = \int \sqrt{9t^4 + 144t^2}\,(dt)$

OE

A1

$s = \int_0^3 3t\sqrt{t^2 + 16}\,dt$

A = 16

A1cso

4

(b) $k(t^2 + A)^{\frac{3}{2}}$

where k is a constant; ft their A

M1

$(t^2 + 16)^{\frac{3}{2}}$

A1

$25^{\frac{3}{2}} - 16^{\frac{3}{2}}$

F(3) − F(0)

m1

= 61

AG

A1 cso

4

[8]

Q2.

(a) $\frac{dy}{dx} = \frac{1}{2\tanh x}$

B1

× sech2*x*

B1

$= \frac{1}{2\sinh x \cosh x}$

for expressing in terms of sinh x and cosh x

M1

$= \frac{1}{\sinh 2x}$

AG; PI by previous line

A1

4

(b) $\sqrt{1+\left(\frac{dy}{dx}\right)^2} = \sqrt{1+\frac{1}{\sinh^2 2x}}$

use of formula; accept √ inserted at any stage

M1

$= \sqrt{\frac{\cosh^2 2x}{\sinh^2 2x}}$

relevant use of cosh2 − sinh2 = 1

M1

$= \frac{\cosh 2x}{\sinh 2x}$

OE

A1

Integral is $\frac{1}{2}$ ln sinh 2*x*

M1 for ln sinh

M1A1

sinh(2ln 4) = $\frac{255}{32}$ sinh(2ln 2) = $\frac{15}{8}$

PI

B1B1

$s = \frac{1}{2}\ln\left(\frac{17}{4}\right)$

ft error in $\frac{1}{2}$

A1F

8

[12]

Q3.

(a) $\frac{dx}{dt} = \sec t - \cos t$

use of FB for sec t;
if done from first principles, allow B1 when
sec t is arrived at

B1,B1

Use of 1 – cos2 t = sin2 t

M1

$\frac{dx}{dt} = \sin t \tan t$

AG

A1

4

(b) $\dot{x}^2 + \dot{y}^2$ = sin2t tan2t + sin2 t

sign error in $\frac{dy}{dt}$ A0

M1A1

Use of 1 + tan2t = sec2 t

m1

$\sqrt{\dot{x}^2 + \dot{y}^2} = \tan t$

ft sign error in $\frac{dy}{dt}$

A1F

$\int_0^{\frac{\pi}{3}} \tan t \, dt = [\ln \sec t]_0^{\frac{\pi}{3}}$

ft sign error in $\frac{dy}{dt}$

A1F

$= \ln 2$

CAo

A1 6

[10]

Q4.

(a) (i) Divide $\cosh^2 t - \sinh^2 t = 1$ by $\cosh^2 t$

Or $\frac{\sinh^2 t}{\cosh^2 t} + \frac{1}{\cosh^2 t}$

M1

Rearrange

AG If solved back to front with no conclusion ending $\cosh^2 t - \sinh^2 t = 1$
B1 only

A1 2

(ii) $\frac{d}{dt}\left(\frac{\sinh t}{\cosh t}\right) = \frac{\cosh^2 t - \sinh^2 t}{\cosh^2 t}$

M1A1

$= \text{sech}^2 t$

AG

A1 3

(iii) $\frac{d}{dt}(\text{sech}\, t) = -(\cosh t)^{-2} \sinh t$

Allow A1 if negative sign missing

M1A1

$= -\text{sech}\, t \tanh t$

AG

A1 3

(b) (i) $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = \text{sech}^4 t + \text{sech}^2 t \tanh^2 t$

Allow slips of sign before squaring for this M1

M1

Use of tanh2 t + sech2 t = 1

Correct formula only for m1

m1

= sech2t

A1

$$\therefore s = \int_0^{\frac{1}{2}\ln 3} \operatorname{sech} t \, dt$$

AG (including limits)

A1 4

(ii) u = et du = et dt

B1

$$\int \operatorname{sech} t \, dt = \int \frac{2}{u^2 + 1} \, du$$

CAO M1 for putting integrand
in terms of u (no sech (ln u))

M1A1

[2 tan−1U]

Or 2 tan−1 et

A1

Change limits correctly or change back to t

At some stage

m1

$$= \frac{2\pi}{3} - \frac{2\pi}{4} = \frac{\pi}{6}$$

CAO

A1 6

[18]

Q5.

(a) $$\frac{d}{dx}\left(\cosh^{-1}\frac{1}{x}\right) = \frac{1}{\sqrt{\frac{1}{x^2} - 1}}\left(-\frac{1}{x^2}\right)$$

M0 if $\frac{dy}{dx} = f(y)$ *and no attempt to substitute back to x*

M1A1

$= \frac{-1}{x\sqrt{1-x^2}}$

AG

A1

3

(b) (i) $\frac{d}{dx}\left(\sqrt{1-x^2}\right) = \frac{-2x}{2\sqrt{1-x^2}}$

For numerator

B1

For denominator (not $(1-x^2)^{\frac{1}{2}}$*)*

B1

$\frac{dy}{dx} = \frac{-x}{\sqrt{1-x^2}} + \frac{1}{x\sqrt{1-x^2}}$

$= \frac{1-x^2}{x\sqrt{1-x^2}} = \frac{\sqrt{1-x^2}}{x}$

For attempt to put over a common denominator

M1

AG

A1

4

(ii) $s = \int_{\frac{1}{4}}^{\frac{3}{4}} \sqrt{1 + \frac{1-x^2}{x^2}}\, dx = \int_{\frac{1}{4}}^{\frac{3}{4}} \frac{1}{x}\, dx$

For use of $\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

M1A1A1

$= [\ln x]_{\frac{1}{4}}^{\frac{3}{4}}$

M1

$= \ln\frac{3}{4} - \ln\frac{1}{4} = \ln 3$

AG

A1

5

[12]

Q6.

(a) (i) $\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + \left(\frac{s}{2}\right)^2}$

Allow M1 for $s = \int \sqrt{1 + \left(\frac{s}{2}\right)^2}\, dx$

then A1 for $\frac{dy}{dx}$

M1A1

$= \frac{1}{2}\sqrt{4 + s^2}$

AG

A1

3

(ii) $\int \frac{ds}{\sqrt{4 + s^2}} = \int \frac{1}{2}\, dx$

For separation of variables; allow without integral sign

M1

$\sinh^{-1}\frac{s}{2} = \frac{1}{2}x + C$

Allow if C is missing

A1

$C = 0$

A1

$s = 2\sinh\frac{1}{2}x$

AG if C not mentioned allow $\frac{3}{4}$

SC incomplete proof of (a)(ii), differentiating

$s = 2\sinh\frac{x}{2}$ *to arrive at* $\frac{ds}{dx} = \frac{1}{2}\sqrt{4+s^2}$

allow M1A1 only $\left(\frac{2}{4}\right)$

A1 4

(iii) $\frac{dy}{dx} = \sinh\frac{1}{2}x$

M1

$y = 2\cosh\frac{1}{2}x + C$

Allow if C *is missing*

A1

$C = 0$

Must be shown to be zero and CAO

A1 3

(b) $y^2 = 4\left(1 + \sinh^2\frac{x}{2}\right)$

Use of cosh2 = 1 + sinh2

M1

$= 4 + s$

AG

A1 2

[12]

Q7.

(a) $\frac{dy}{dx} = \frac{1}{\tanh\frac{x}{2}}$...

B1

$\operatorname{sech}^2\frac{x}{2}$...

B1

$\frac{1}{2}$

B1

$= \dfrac{1}{2\dfrac{\sinh\frac{x}{2}}{\cosh\frac{x}{2}}\cosh^2\dfrac{x}{2}}$

OE ie expressing in sinh $\frac{x}{2}$ *and cosh* $\frac{x}{2}$

M1

$= \dfrac{1}{2\sinh\dfrac{x}{2}\cosh\dfrac{x}{2}}$

$= \dfrac{1}{\sinh x}$

ie useof sinh 2A $= 2\sinh A\cosh A$

M1

= cosech x

AG

A1

Alternative

$\ln\sinh\dfrac{x}{2} - \ln\cosh\dfrac{x}{2}$

(B1)

$\dfrac{1}{2}\dfrac{\cosh\dfrac{x}{2}}{\sinh\dfrac{x}{2}} - \dfrac{1}{2}\dfrac{\sinh\dfrac{x}{2}}{\cosh\dfrac{x}{2}}$

(B1B1)

$\dfrac{\cosh^2\dfrac{x}{2} - \sinh^2\dfrac{x}{2}}{2\sinh\dfrac{x}{2}\cosh\dfrac{x}{2}}$

(M1)

Use of sinh2A = 2 sinh A cosh A

(M1)

result (A1) 6

(b) (i) $s = \int_1^2 \sqrt{1 + \text{cosech}^2 x}\, dx$ M1

$= \int_1^2 \coth x\, dx$

AG A1 2

(ii) $s = [\ln \sinh x]_1^2$

needs tobe correct M1

= ln sinh 2 – ln sinh 1 A1

$= \ln \dfrac{2 \sinh 1 \cosh 1}{\sinh 1}$

must be seen A1F

= ln (2 cosh 1)

AG A1 4

[12]

Q8.

(a) (i) $\dfrac{d}{dt}\left(\dfrac{1}{\cosh t}\right) = -1(\cosh t)^{-2} \sinh t$

Or $\dfrac{-2(e^t - e^{-t})}{(e^t + e^{-t})^2}$ M1A1

= −sech *t* tanh *t*

AG A1 3

(ii) Use of tanh2 t = 1 sech2 t

M1

Printed result

A1

2

(b) (i) $\dot{x} = 1 - \operatorname{sech}^2 t \quad (\dot{y} = -\operatorname{sech} t \tanh t)$

B1

$\dot{x}^2 + \dot{y}^2 = (1 - \operatorname{sech}^2 t)^2 + \operatorname{sech}^2 t - \operatorname{sech}^4 t$

Any form

M1A1

= 1 – sech2 t= tanh2t

AG

A1

4

(ii) $s = \int_0^t \tanh t \, dt$

Ignore limits for M1 and first A1

M1

$= [\ln \cosh t]_0^t$

A1

$= \ln \cosh t$

AG

A1

3

(iii) es cosh t

M1

y = e–s

AG

A1

2

(c) $S = 2\pi \int_0^t \operatorname{sech} t \tanh t \, dt$

Ignore limitsfor M1 and first A1

M1

$= 2\pi\left[-\operatorname{sech} t\right]_0^t$

A1

$= 2\pi(1 - \operatorname{sech} t)$

A1

$= 2\pi(1 - e^{-s})$

AG

A1 4

[18]

Q9.

(a) $\frac{dy}{dx} = \frac{2}{\sqrt{x}}$

accept $2x^{-\frac{1}{2}}$ *etc*

B1

$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + \frac{4}{x}}$

ft sign error in $\frac{dy}{dx}$

M1A1F

$= \sqrt{\frac{x+4}{x}}$

AG

A1 4

(b) (i) $x = 4\sinh^2\theta$, $dx = 8\sinh\theta\cosh\theta\, d\theta$

M1 for any attempt at $\frac{dy}{d\theta}$

M1A1

$I = \int \sqrt{\frac{4\sinh^2\theta + 4}{4\sinh^2\theta}}\ 8\sinh\theta\cosh\theta\, d\theta$

M1

$= \int \frac{2\cosh\theta}{2\sinh\theta} 8 \sinh\theta \cosh\theta \, d\theta$

ie use of $\cosh^2\theta - \sinh^2\theta = 1$

m1

$= \int 8\cosh^2\theta \, d\theta$

AG

A1

5

(ii) Use of $2\cosh^2\theta = 1 + \cosh 2\theta$

allow if sign error

M1

$I = \int 4(1+\cosh 2\theta)\, d\theta$

oe

A1

$= 4\theta + 2\sinh 2\theta$

oe

A1F

Use of $\sinh 2\theta = 2\sinh\theta\cosh\theta$

m1

$= 4\sinh^{-1}\frac{1}{2} + 4 \times \frac{1}{2}\sqrt{1+\frac{1}{4}}$

A1F

$= 4\sinh^{-1}\frac{1}{2} + \sqrt{5}$

AG

A1

6

[15]

Q10.

(a) (i) $2\left(\frac{e^{\theta} - e^{-\theta}}{2}\right)\left(\frac{e^{\theta} + e^{-\theta}}{2}\right) = \frac{e^{2\theta} - e^{-2\theta}}{2} = \sinh 2\theta$

AG

M1A1

2

(ii) $\left(\frac{e^{\theta}-e^{-\theta}}{2}\right)^2+\left(\frac{e^{\theta}+e^{-\theta}}{2}\right)^2$

M1

$= \frac{e^{2\theta}-2+e^{-2\theta}+e^{2\theta}+2+e^{-2\theta}}{4}$

A1

$= \cosh 2\theta$

AG

A1

3

(b) (i) $\dot{x} = 3\cosh 2\theta \sinh \theta$

Allow M1 for reasonable attempt at differentiation,

but M0 for putting in terms of ekθ or sinh 3θ

unless real progress made towards $\dot{x}^2 + \dot{y}^2$

M1

$\dot{y} = 3\sinh 2\theta \cosh \theta$

Allow this M1 if not squared out, must be clear

sum in question is $\dot{x}^2 + \dot{y}^2$

A1

$\dot{x}^2 + \dot{y}^2 = 9\cosh 4\theta \sinh 2\theta + 9\sinh 4\theta \cosh 2\theta$

AG

M1

$= 9\sinh 2\theta \cosh^2 \theta (\cosh 2\theta + \sinh 2\theta)$

Accept $\int_0^1 \sqrt{\frac{9}{4}\sinh^2 2\theta \cosh 2\theta}\, d\theta$

but limits must appear somewhere

A1

$= \frac{9}{4}$ sinh2 2θ cosh θ2

AG

A1

6

(ii) $S = \int_0^1 \frac{3}{2} \sinh 2\theta \sqrt{\cosh 2\theta}\, d\theta$

M1

$u = \cosh 2\theta \qquad du = 2\sinh 2\theta\, d\theta$

M1A1

$I = \int \frac{3}{4} u^{\frac{1}{2}} du = \frac{3}{4} \times \frac{2}{3} u^{\frac{3}{2}}$

A1F

$S = \left\{ \frac{1}{2} (\cosh 2\theta)^{\frac{3}{2}} \right\}_0^1$

A1F

$= \frac{1}{2} \left\{ (\cosh 2)^{\frac{3}{2}} - 1 \right\}$

AG

A1

6

[17]

Examiner reports

Q1.

(a) Most students were able to establish the given answer for the arc length, although some neglected to insert the limits and others omitted the dt on the integral and so lost the final mark. Others were unable to find the correct value of A when trying to simplify an expression inside the square root sign.

(b) Students resorted to a variety of substitutions in order to find the value of the integral. Quite a few used hyperbolic functions but were then unable to integrate their resulting expression of the form $k\cosh^2\theta\sinh\theta$; others used a substitution such as $v^2 = t^2 + 16$ quite effectively. However, many students did not change their limits when making a substitution, writing expressions such as $\int_0^3 \frac{3}{2}u^{\frac{1}{2}}$ du and this was penalised.

It had been expected that most students would have been able to write down the answer to the integral as $(t^2 + 16)^{\frac{3}{2}}$ before evaluating the limits. It was then necessary to show some working such as $s = 125 - 64 = 61$, since the numerical value of the answer was given in the question.

Q2.

Overall, responses to this question were good and there were many completely correct or nearly correct solutions. Part (a) was usually correct, although lack of intermediate working made it difficult to decide whether a candidate really saw what to do, or merely wrote down the printed answer.

In part (b) the commonest errors were to either think that $\sqrt{(1 + 1/\sin^2 2x)}$ was equal to $1 + 1/\sin 2x$, or to lose the $2x$ in the process of simplification and end up with $\coth x$ instead of $\coth 2x$. Many candidates arriving at $\int \coth 2x\,dx$ were able to perform the integration correctly, although a few lost the factor ½.

Q3.

Part (a) was extremely badly done, with few candidates scoring more than one out of the four available marks. The reason for this was that candidates failed to realise that the derivative of ln (sec t + tan t) is given in the formulae booklet, and so, in attempting to do the differentiation themselves, they became bogged down with the manipulation of trigonometrical functions. Part (b) was better attempted and there were many correct solutions, but, if a solution did peter out, it was generally at the stage where the candidate obtained sin t sec t but failed to realise that that was in fact tan t.

Q4.

Part (a) was a source of good marks for almost all candidates. If errors did occur they were usually errors of sign. Part (b) was also generally well done although it was disappointing to see the square root of $\operatorname{sech}^2 t\tanh^2 t + \operatorname{sech}^4 t$ written as $\operatorname{sech} t\tanh t + \operatorname{sech}^2 t$ a significant number of times. Responses to part (b)(ii) were mixed. Poor algebraic manipulation in the handling of sech t when expressed in terms of u let many candidates down badly so that they ended up with a polynomial in u to integrate.

Q5.

Few candidates were able to supply a correct proof of part (a). The derivative of $\cosh^{-1}\left(\frac{1}{x}\right)$ was almost invariably given as $\frac{1}{\sqrt{\left(\frac{1}{x}\right)^2 - 1}}$ (followed by the correct answer after a small amount of spurious algebra. Candidates fared better in part (b)(i), although it was a little surprising how many candidates were unable to differentiate $\sqrt{1-x^2}$ correctly. Those candidates who continued and attempted part (b)(ii) frequently produced a correct solution. Solutions to this part which went awry were those in which candidates wrote $\sqrt{1+\left(\frac{\sqrt{1-x^2}}{x}\right)^2}$ followed by $\sqrt{1+\frac{1-x}{x}}$ (which incidentally gave the correct answer!), or in some cases where candidates arrived at $\sqrt{\frac{1}{x^2}}$ or $\sqrt{x^{-2}}$ but were unable to take the square root correctly.

Q6.

Responses to part (a)(i) of this question were reasonable, although candidates starting with $s = \int\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$ tended to flounder. Part (a)(ii) was very poorly attempted with the majority of candidates failing to realise that the way forward was to separate the variables in order to integrate. It was very common to see attempts at $\int\sqrt{4+s^2}\,dx$ treated as if it were $\int\sqrt{4+s^2}\,ds$.

Of the few that did manage to separate the variables, virtually no one considered the boundary conditions but merely assumed that the constant of integration was zero. Candidates were more successful with part (a)(iii) and, although the constant of integration was omitted in many cases, more candidates considered the boundary conditions than in part (a)(ii).

Those candidates who managed part (a)(iii) usually went on to work part (b) correctly.

Q7.

Although many candidates were able to write down $\frac{1}{\tanh\frac{x}{2}}$ multiplied by $\frac{1}{2}\operatorname{sech}^2\frac{x}{2}$, fewer were able to combine these results to obtain cosech x. Even those candidates who expressed $\frac{dy}{dx}$ entirely in terms of $\cosh\frac{x}{2}$ and $\sinh\frac{x}{2}$ seemed to baulk at the algebra which led to $\frac{1}{2\sinh\frac{x}{2}\cosh\frac{x}{2}}$.

Part (b)(i) was done well and many candidates were able to arrive at $s = \ln\sinh 2 - \ln\sinh 1$ in part (b)(ii) but were unable to reach the printed answer. If the integral of coth x was performed incorrectly, it was often by coth x being replaced by $\frac{1}{\tanh x}$ followed by ln tanh x or ln cosh x as the integral.

Q8.

Generally, candidates scored quite well on this question. Some candidates struggled with part (a)(i) by not realising that sech t was $(\cosh t)^{-1}$, instead expressing sech t in exponential form.

This latter method rarely led to a correct solution. However, apart from some sign fudging in part (a)(ii), most candidates were able to recover to answer parts (a)(ii) and (b)(i) correctly. Very few candidates were able to score the three available marks in part (b)(ii), by either ignoring the limits of integration completely or by writing $s = \ln\cosh t + c$ with no effort to show that the value of c was zero. In spite of some inelegant methods, part (b)(iii) was usually answered correctly.

Part(c) caused problems, often by candidates attempting to integrate $e^{-s}\tanh t$ with respect to t by regarding e^{-s} either as a constant or else as e^{-t}. Of those who correctly integrated to arrive at –sech t, again very few candidates recognised the need for limits and so were unable to arrive at the printed result.

Q9.

This question was generally answered well and many candidates were able to score 12 out of the available 15 marks. Part (a) was well answered apart from a few candidates who wrote $\frac{dy}{dx} = 2x^{-\frac{1}{2}}$ dy followed by $\frac{1}{2x^{\frac{1}{2}}}$. In part (b) there were two main sources of error. The first was to interchange dx with dθ without any consideration of $\frac{dx}{d\theta}$, and the second was to write $\sqrt{\frac{4\sinh^2\theta + 4}{4\sinh^2\theta}}$ as $\frac{2\sinh\theta + 2}{2\sinh\theta}$. There were also a few candidates who were unable to differentiate $4\sinh^2\theta$.

In part (b)(i), most candidates were able to integrate $8\cosh^2\theta$ correctly but few were able to arrive at the printed result in part (b)(ii). Two factors contributed to this. Candidates either failed to change the limits for x to the corresponding limits for θ or else wrote the answer with no evident method. This was unacceptable as the answer for the arc lengths was given.

Q10.

Candidates were generally well drilled in proving the identities of parts (a)(i) and (a)(ii) although in (a)(ii) sometimes $(e^{\theta} - e^{-\theta})^2$ was written as $e^{2\theta} + e^{-2\theta}$ with the same result from $(e^{\theta} + e^{-\theta})^2$, thus obtaining the correct answer from incorrect algebra. Part (b)(i) was usually quite well done so long as candidates did not write $\cosh^3\theta$ as $\frac{1}{8}(e^{\theta} + e^{-\theta})^3$.

Those who worked in powers of e^{θ} and $e^{-\theta}$ found it impossible to reconcile their formula with the printed result and so make little meaningful progress. Part (b)(ii) proved to be beyond all but the most able candidates. The required integral, $\int \frac{3}{2} \sinh 2\theta \sqrt{\cosh 2\theta} \, d\theta$, was tackled successfully by these candidates by a variety of methods. Some spotted the integral, others used the substitution $u = \cosh 2\theta$ whilst others again integrated by parts.